

Send: 01 10 00 D6 00 01 02 00 00 (sets DI1 logic active low)

Setting Multi-Segment Position/Speed Enable: 2003_h :05_h (00D7_h)=28, Logic is active low:
2003_h :06_h (00D8_h)=0;

Send: 01 10 00 D7 00 01 02 00 1C (sets DI2 to multisegment position enable)

Send: 01 10 00 D8 00 01 02 00 00 (sets DI2 logic active low)

- run

Motor enable: DI1 input low;

Multi-segment position/speed enable, DI2 input low, the motor runs in accordance with the operation mode and operation parameters set in the 2011_h (0291_h ~02E6_h) Multi-segment position function parameter set.

5.17 Multi-Speed Mode

5.17.1 Structure Plan

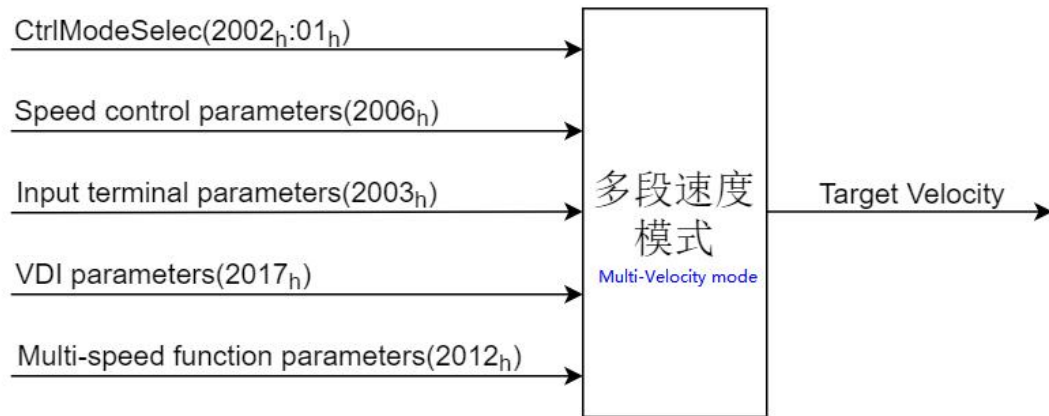


Figure 5- 36

5.17.2 Relevant Object

The following objects require attention in this mode:

Table 5- 40

Object Index	Description
2002 _h : 01 _h (00B1) _h	Operation mode selection
2006 _h : 01 _h (0148) _h	NiMotion speed mode master speed feed source, 0-digital feed, 3-duty cycle feed (20Hz-20kHz, 1kHz recommended)
2006 _h : 02 _h (0149) _h	NiMotion speed mode auxiliary speed feed source, 0 - digital feed, 3 - multispeed
2006 _h : 03 _h (014A) _h	NiMotion speed mode speed command selection, 0 - primary speed given, 1 - secondary speed given, 2 - primary and secondary speed given (default 0)
2006 _h : 04 _h (014B) _h	Speed command numerical given value in rpm
2006 _h : 07 _h (014E) _h	The speed command acceleration ramp time constant, in ms, indicates the time taken to accelerate from 0 rpm to 1000 rpm.
2006 _h : 08 _h (014F) _h	Speed command deceleration ramp time constant in ms
2017 _h (0326 _h ~0346) _h	Functional configuration of the virtual input terminals (see: "6.1.1 Digital input")
2003 _h (00D5 _h ~00E6) _h	Functional configuration of the physical input terminals (see: "6.1.1 Digital input") Note: You cannot have multiple input terminals configured as the same switch at the same time, or you will report a terminal setup fault.
2012 _h (02E9 _h ~0323) _h	Multi-Segment Speed Function Parameter Set

5.17.3 Control Commands and Status Messages

There is no state machine management in this mode, the motor is enabled via the terminals.

Enable

Prerequisite for operation in this mode: Motor operation mode is switched to NiMotion speed mode (2002_h : 01_h (00B1_h) set to 2).

1. Speed source selection:

The object is the auxiliary speed command source 2006_h :02_h (0149_h), meaning: 0-number given, 3-multiple speeds,;

With the speed command selection 2006_h :03_h (014A_h) it is the main speed command, the auxiliary speed command or the main and auxiliary speed commands that actually take effect.

2. Set the multi-function terminals (both physical and virtual terminals are applicable, see "6.1 Digital Inputs and Outputs" for details on terminal settings):

Configure one input terminal as motor enable, and configure its valid logic level according to the actual situation, the default low level is valid.

Configure the other input terminal as multi-segment enable (function No. 28), and configure its effective logic level according to the actual situation, the default is effective low level.

3. Speed Setting

Setting of the corresponding operating mode and operating parameters for each segment via the multi-segment speed function parameter group 2012_h (02E9_h ~0323_h)

4. (of a computer) run

If the motor is enabled via one of the terminals, the motor operates in accordance with the operation mode set in the multispeed function parameter set 2012_h (02E9_h ~0323_h) and the operation parameters for each segment.

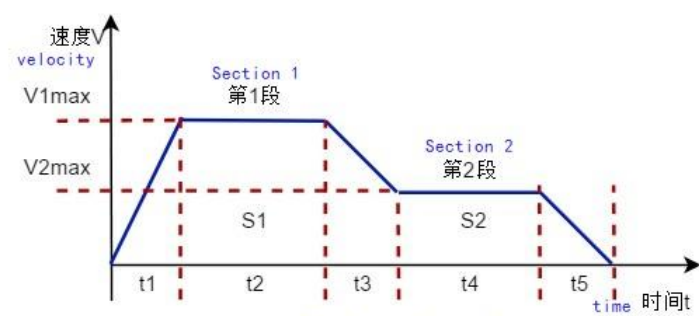
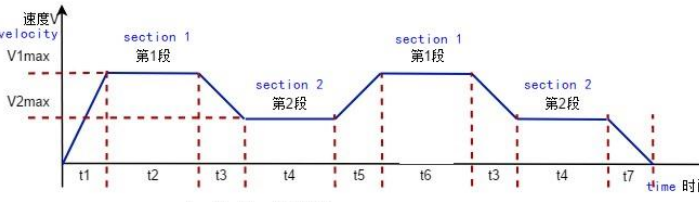
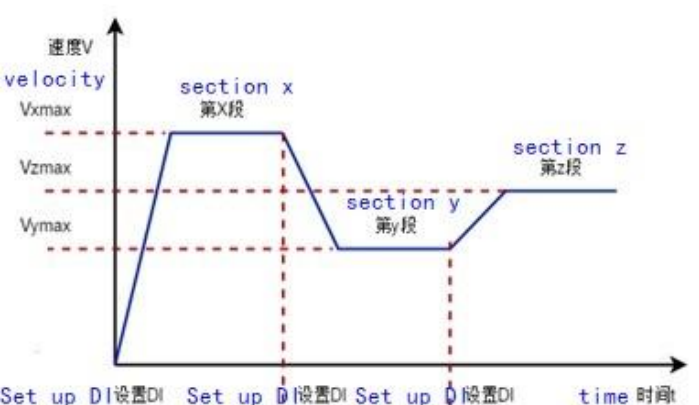
5.17.4 Functional Description

The multi-segment speed mode has 3 types of operation, namely single run, cycle run and DI switching run. When using multi-segment position/speed function, one DI port must be set as motor enable (function No. 1) and multi-segment enable (function No. 28); during the operation of each segment speed command, it must be ensured that the motor enable is valid, if the motor enable is turned off, the motor will be shut down according to the deactivation mode. In the non-DI switching operation mode, in addition to the motor enable being valid, the multi-segment enable must be guaranteed to be valid; if the multi-segment enable is turned off, the motor will not be able to switch between multi-segment speeds.

Specific descriptions of each mode of operation are shown in the table below:

Table 5-41

Operation mode	Description	Running curve
----------------	-------------	---------------

Operation mode	Description	Running curve					
single run 2012 _h :01 _h (02E9 _h) = 0	Paragraphs 1 through 16 were run once; Automatic incremental switching of segment numbers. Multi-segment enable is level active.	 The first and second section speed V_{1max}、V_{2max}: 第1段、第2段指令速度 第1段运行时间为 t₁+t₂, 第2段运行时间为 t₃+t₄, 以此类推 The first run time is T₁ + T₂, the second run time is T₃ + T₄, and so on					
run in a loop 2012 _h :01 _h (02E9 _h) = 1	Segment 1 to the end segment set by 2012 _h :02 _h (02EA _h); Run 2012 _h :03 _h (02EB _h) the number of times set; Automatic incremental switching of segment numbers. Multi-segment enable is level active.	 V_{1max}、V_{2max}: 第1段、第2段指令速度 The first and second instruction speed 第1段运行时间为 t₁+t₂, 第2段运行时间为 t₃+t₄, 以此类推 The first segment runs for t₁ + t₂, the second segment runs for t₃ + t₄, and so on					
DI switching operation 2012 _h :01 _h (02E9 _h) = 2 Note: Physical and virtual terminals can be used in combination and the combination must be 4-digit; for example: Physical terminal selection DI1=6,DI2=7,Virtual terminal VI1=8,VI2=9	The speed segment number switching is determined by the decimal 0 to 15 corresponding to the 4-digit binary number consisting of multifunction numbers 6, 7, 8 and 9; Enabling is valid for continuous operation; Each speed command run time is determined only by the speed segment number switching interval. Multi-segment enable is active along the change.	 Set up DI设置DI Set up DI设置DI Set up DI设置DI V_{xmax}、V_{ymax}、V_{zmax}: 第x段、第y段、第z段指令速度 Section x, y, Z instruction speed					
	Correspondence table	<table><tr><td>Fun9</td><td>Fun8</td><td>Fun7</td><td>Fun6</td><td>section</td></tr></table>	Fun9	Fun8	Fun7	Fun6	section
Fun9	Fun8	Fun7	Fun6	section			

Operation mode	Description	Running curve					
	between segment number and multifunction number					number	
		0	0	0	0	1	
		0	0	0	1	2	
							
		1	1	1	1	16	

5.17.5 Configuration Example

Multi-Speed-Single Run Control

- Parameter Settings:

Set to NiMotion speed mode 2002_h:01_h (00B1_h)=2;

Send: 01 10 00 B1 00 01 02 00 02

Set the speed command source to multispeed: 2006_h:02_h (0149_h)=3, 2006_h:03_h (014A_h)=1;

Send: 01 10 01 49 00 01 02 00 03 (set speed command source to multispeed)

Send : 01 10 01 4A 00 01 02 00 01 (set speed source to auxiliary speed command source)

Sets the multispeed function parameter group:

Multi-speed operation mode 2012_h:01_h (02E9_h) = 0 (single run);

Send: 01 10 02 E9 00 01 02 00 00

The operating parameters for the 1st speed:

Speed command 2012_h:0C_h (02F4_h) = 500 (in rpm), command runtime 2012_h:0D_h (02F5_h) = 2000 (in ms), the

Send: 01 10 02 F4 00 01 02 01 F4 (sets first segment speed to 500 rpm)

Send: 01 10 02 F5 00 01 02 07 D0 (set first speed run time to 2000ms)

Acceleration and deceleration time 2012_h:0E_h (02F6_h) = 500 (in ms);

Send: 01 10 02 F6 00 01 02 01 F4

The operating parameters for segments 2 through 16 are set with reference to the operating parameters for segment 1 speeds;

Set motor enable: 2003_h:03_h (00D5_h)=1, logic active low: 2003_h:04_h (00D6_h)=0;

Send: 01 10 00 D5 00 01 02 00 01 (set DI1 to enable)

Send: 01 10 00 D6 00 01 02 00 00 (sets DI1 logic active low)

Setting Multi-Segment Position/Speed Enable: 2003_h:05_h (00D7_h)=28, Logic is active low: 2003_h:06_h (00D8_h)=0;

Send: 01 10 00 D7 00 01 02 00 1C (sets DI2 to multisegment position enable)

Send: 01 10 00 D8 00 01 02 00 00 (sets DI2 logic active low)

- run

Motor enable: DI1 input low;

Multi-segment position/speed enable, DI2 input low, the motor runs in accordance with the operation mode set in the 2012_h (02E9_h~0323_h) Multi-Segment Speed Function Parameter Set and the operation parameters of each segment.

6 Special Features

6.1 Digital Inputs and Outputs

The integrated low-voltage servo motors have input and output interfaces. The integrated digital input function is used to connect sensors for auxiliary motion control, such as limit microswitches, photoelectric switches, Hall sensors, etc. The digital output function can output 8mA current to drive solid state relays or indicators. The digital output function can output 8mA current to drive solid state relays or indicator lights etc.

6.1.1 Digital Inputs

Related Object Settings

Table 6- 1

Object Index	Description
2003 _h (00D5 _h ~00E6 _h)	Physical Input Terminal Function Configuration
2017 _h (0326 _h ~0346 _h)	Virtual Input Terminal Function Configuration
2004 _h (00F8 _h ~0101 _h)	DX1 terminal function selection
200C _h : 07 _h (0235 _h)	Virtual digital input enable, 0-disable, 1-enable
2031 _h : 01 _h (0373 _h)	Communication Given VDI Virtual Levels
200B _h : 05 _h (01E2 _h)	Actual level status corresponding to entity terminals (bit0-DI1~bit3-DI4)

control command

This device has up to 3 digital input entity terminals: fixed input DI1~3;

Communication virtual input terminals, set for use via the 2017_h (0326_h ~ 0346_h) parameter set. 2031_h : 01_h (0373_h) sets the state of the corresponding level of the virtual terminal for communication (bit0-VDI1 ~ bit15-VDI16).

6.1.1.1 Physical Terminal

Special Function Options

The specific values are taken from the table below:

Table 6- 2

function number	functional definition
0	undefined
1	enable
2	Alarm reset
6	Multi-segment operation command switching1
7	Multi-segment operation command switching 2
8	Multi-segment operation command switching3
9	Multi-segment operation command switching4
12	pause (media player)
14	Positive limit switch
15	Reverse limit switch
20	Step amount enable
21	Nimotion Speed Mode RPM Direction Commands
28	Multi-segment position/speed enable
31	Home switch
33	Setting the origin
36	Analog input position/speed control direction
38	Clearing Fault History
39	Clear power-up time
40	Quadrature pulse input A (fixed DI1)
41	Quadrature pulse input B (fixed DI2)
42	Pulse input (fixed DI1)
43	Pulse input direction (fixed DI2)
44	Duty cycle input (fixed DI1)
45	Duty cycle input direction (fixed DI2)

Terminal Logic Selection

Multi-function terminal logic selection is used through the 2003_h (00D5_h ~00E6_h) (Entity Input Terminal) parameter group setting. The specific values are shown in the table below:

Table 6- 3

Set Value	Logical Definition
0	Active Low
1	Active High
2	Rising edge valid
3	Effective along the

	descending edge
4	Valid on both rising and falling edges

Notes:

- 1、 "1-enable" is only valid in NiMotion mode, which is equivalent to writing enable control word to 6040_h (0380_h) in 402 mode.
- 2, different terminals can not be set the same function number, if set incorrectly will report parameter error fault.
3. If the working mode is set to HM mode of CiA402, without setting 14, 15 and 31 function numbers will also report parameter failure.
4. Terminal function and logic selection are set to shutdown effective.
5. The digital inputs are detected only once every millisecond. Cannot handle signal changes of less than 1 ms at the input.

example

The following configuration of the entity input terminal DI1 is a positive limit switch

- Configure DI1 as a positive limit switch, 2003_h : 03_h (00D5_h)=14
Send: 01 10 00 D5 00 01 02 00 0E
- Logic selection is active low, 2003_h : 04_h (00D6_h) =0
Send: 01 10 00 D6 00 01 02 00 00

6.1.1.2 Virtual Terminal

Virtual VDI Terminal Special Function Definition

The specific values are taken from the table below:

Table 6- 4

Function Number	Functional Definition
0	undefined
1	enable
2	Alarm reset
6	Multi-segment operation command switching1
7	Multi-segment operation command switching 2
8	Multi-segment operation command switching3
9	Multi-segment operation command switching4
12	pause (media player)
14	Positive limit switch
15	Reverse limit switch
20	Step amount enable
21	Nimotion Speed Mode RPM Direction Commands
28	Multi-segment position/speed enable
31	Home switch

Function Number	Functional Definition
33	Setting the zero point
36	Analog input position/speed control direction
38	Clearing Fault History
39	Clear power-up time

Notes:

1、"1-Enable""12-Suspend "is only valid in NiMotion mode, which is equivalent to writing enable control word to 6040_h (0380_h) in 402 mode.

Virtual VDI Terminal Logic Selection

Multi-function terminal logic selection is used through the 2017_h (0326_h ~0346_h) (Virtual Input Terminal) parameter group setting. The specific values are shown in the table below:

Table 6- 5

Preset Value	Logic Definition
0	Active High
1	Rising edge valid

start

The Integrated Low Voltage Servo Motor Virtual VDI Terminal is functionally active when index 200C_h : 07_h (0235_h) is 1, and a value of 0 for 200C_h : 07_h (0235_h) disables the Virtual DI Terminal function. Enable is performed by entering the terminal function number and logic selection via the 2017_h (0326_h ~0346_h) sub-index, and when the logic selection is valid, the function of the corresponding function number is valid. Index 2031_h : 01_h (0373_h) is the state corresponding to the virtual input terminal (bit0-VDI1 ~ bit15-VDI16), which can only be given by communication, as defined below:

2031_h : 01_h (0373_h)

Bit	15-7	6	5	4	3	2	1	0
Description	VDI16-VDI8	VDI7	VDI6	VDI5	VDI4	VDI3	VDI2	VDI1

Example

The following virtual input terminal DI1 is configured as a positive limit switch

- Configure DI1 as a positive limit switch, 2003_h : 03_h (00D5_h)=14
Send: 01 10 00 D5 00 01 02 00 0E
- Logic selection is active low, 2003_h : 04_h (00D6_h) =0
Send: 01 10 00 D6 00 01 02 00 00

6.1.2 Digital Outputs
Related Object Settings

Table 6- 6

2004 _h (00F8 _h ~0101 _h)	Output Terminal Configuration Parameters
2031 _h : 02 _h (0374 _h)	Communication given VD0 output state (bit0~DO)

control command

This device has 2 digital output entity terminals, which need to be configured with the object 2004_h (00F8_h ~0101_h) to select the output terminal function and logic level. The output status can also be given by communication through object 2031_h : 02_h (0374_h).

Output terminal function selection:

Table 6- 7

Function Number	Function Definition
0	Configure DX1 as the input
1	Configure DX1 as an output
2	Motor running stop
3	target met
4	Alarm output
5	Brake output
6	External braking resistor function
30	Brake power saving drive output

Note: One of the DO ports is dedicated to the external brake resistor function and the brake power saving drive output function;

Output terminal logic selection:

Table 6- 8

Set Value	Logical Definition
0	Active Low
1	Active High

give an example

Example 1: Configure DO1 as a normal DO function and output the DO1 port to a conductive state:

- Configure DO1 as an output, 2004_h : 01_h (00F8_h) = 1;
Send: 01 10 00 F8 00 01 02 00 01
- Configure DO1 logic to be active high, 2004_h : 02_h (00F9_h) = 1;
Send: 01 10 00 F9 00 01 02 00 01
- Communication is given that DO1 output is on, 2031_h : 02_h (0374_h) = 1, at this time DO1 to GND output state on.
Send: 01 10 03 74 00 01 02 00 01

6.2 Analog Inputs

The implementation of the analog control mode requires voltage input to the analog input port, and the voltage input range is 0-10V; through the configuration parameters, the range of controlled quantity corresponding to 0-10V is set; the analog input control includes analog speed control, analog torque control and analog position control;

6.2.1 Analog position control

Table 6- 9

Object Index	Description
2002 _h :01 _h (00B1) _h	Operation mode selection
2005 _h :01 _h (010D) _h	Position command source, 3-mode 1 analog position control, 4-mode 2 analog position control
2005 _h :2A _h (0140) _h	AI1 dead-band judgment time window, unit 1ms
2005 _h :2C _h (0142) _h	Analog 10V corresponding position value (encoder unit)
2017 _h (0326 _h ~0346) _h	Functional configuration of the virtual input terminals (see: "6.1.1 Digital input")
2003 _h :17 _h (00E9) _h	AI1 bias in 1mV
2003 _h :18 _h (00EA) _h	AI1 input filter time constant in 1ms
2003 _h :19 _h (00EB) _h	AI1 Dead-band in 0.01mV
2003 _h :1A _h (00EC) _h	AI1 Magnification in 0.001

Enable

There is no state machine management in this mode and the motor is enabled to run via the terminals.

Prerequisite for this mode of operation: the integrated low-voltage servomotor is switched to NiMotion position mode (2002_h:01_h (00B1_h) write 1).

1. Location Source Selection:

Set Position Command Source 2005_h:01_h (010D_h), Meaning: 3-Mode 1 analog position control, 4-Mode 2 analog position control;

2. Setting the multi-function terminals (both physical and virtual terminals are applicable, see "6.1 Digital Inputs and Outputs" for details on terminal settings).

Configure one input terminal as motor enable, and configure its valid logic level according to the actual situation, the default low level is valid;

If the set position command source is 3, the corresponding input terminal DI1 or DI2 needs to be set for analog input position control direction function (multifunction number 36).

3. Position Setting

Configure the analog 10V to correspond to the position value 2005_h:2C_h (0142_h), which is the absolute motor position.

4. run

The motor is enabled via a terminal, and the motor operates in the corresponding position according to the analog input.

Note: The analog position control is an absolute position control and the current position needs to be cleared to 0 before use.

Analog position control mode

There are two modes of analog input position control: mode 1 and mode 2.

1. Model 1:

Given position = Symbol (multifunction number 36 level corresponds to) * Analog 10V corresponds to position value * (Input voltage * AI1 multiplier + AI1 bias)/10000, multifunction number 36 is valid for negative direction, invalid for positive direction.

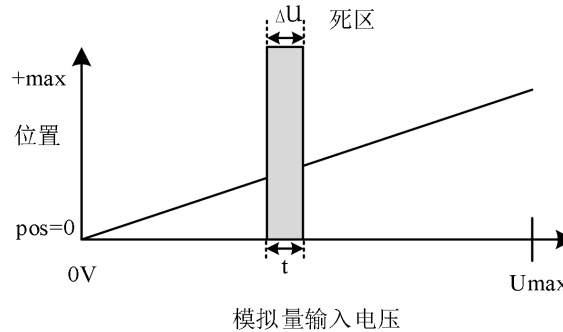


Figure 6- 1

2. Mode 2:

Input voltage 5~10V: given position value = Symbol (multifunction number 36 level corresponds to) * Analog 10V corresponding position value * (Input voltage * AI1 multiplier + AI1 bias)/10000;

Input voltage 0~5V: given position value = -symbol (multifunction number 36 level corresponds to) * analog 10V corresponding position value * (input voltage * AI1 multiplier + AI1 bias)/10000;

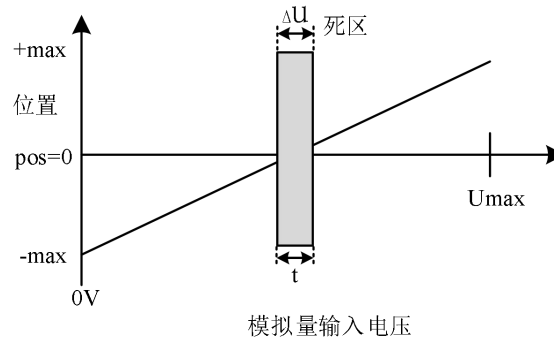


Figure 6- 2

Note: The given position before dead-band is the analog voltage ($\Delta u/t$) after input filtering and multiplicative bias in the AI1 dead-band judgment time window $2005_h:2A_h$ (0140_h) within the change of continuous greater than the AI1 dead-band $2003_h:19_h$ (00EB_h) setting value, otherwise the given position maintains the value of the previous moment, the setting can eliminate the position jitter brought about by the jitter of the analog voltage.

Configuration Example

- Parameters setting

Set to NiMotion position mode: $2002_h:01_h$ (00B1_h)=1;

Send: 01 10 00 B1 00 01 02 00 01

Set the location command source to 3: $2005_h:01_h$ (010D_h)=3;

Send: 01 10 01 0D 00 01 02 00 03

Set motor enable: 2003_h:03_h (00D5_h)=1, logic active low: 2003_h:04_h (00D6_h)=0;

Send: 01 10 00 D5 00 01 02 00 01 (set DI1 to enable)

Send: 01 10 00 D6 00 01 02 00 00 (sets DI1 logic active low)

Set the analog input position control direction: 2003_h:05_h (00D7_h)=36, Logic is falling edge valid: 2003_h:06_h (00D8_h)=3;

Send: 01 10 00 D7 00 01 02 00 24 (set DI2 to analog input position control direction 36)

Send: 01 10 00 D8 00 01 02 00 03 (sets DI2 logic to be active on falling edge)

2005_h:2C_h (0142_h)=10000; (position value corresponding to 10V)

Send: 01 10 01 42 00 02 04 00 00 27 10

The AI interface is connected to the analog voltage value (be careful not to reverse the positive and negative), and the position is given by adjusting the voltage;

- run

DI1 input low level, the motor runs according to the position given by the analog, DI2 controls the direction of motor running.

6.2.2 Analog speed control

relevant object

Table 6- 10

Object Index	Description
2002 _h :01 _h (00B1) _h	Operation mode selection
2006 _h :01 _h (0148) _h	NiMotion speed mode master speed feed source, 1-mode 1 analog speed control
2006 _h :02 _h (0149) _h	NiMotion speed mode auxiliary speed feed source, 1-mode 2 analog speed control
2006 _h :03 _h (014A) _h	NiMotion speed mode speed command selection, 0 - primary speed given, 1 - secondary speed given
2006 _h :04 _h (014B) _h	Speed command numerical given value in rpm
2006 _h :07 _h (014E) _h	The speed command acceleration ramp time constant, in ms, indicates the time taken to accelerate from 0 rpm to 1000 rpm.
2006 _h :08 _h (014F) _h	Speed command deceleration ramp time constant in ms
2006 _h :0A _h (0151) _h	Forward speed threshold in rpm
2006 _h :0B _h (0152) _h	2006 _h :0B _h : Forward speed threshold, in rpm
2017 _h (0326 _h ~0346) _h	Functional configuration of the virtual input terminals (see: "6.1.1 Digital input")
2003 _h :17 _h (00E9) _h	AI1 bias in 1mV
2003 _h :18 _h (00EA) _h	AI1 input filter time constant in 0.01ms
2003 _h :19 _h (00EB) _h	AI1 Dead-band in 1mV
2003 _h :1A _h (00EC) _h	AI1 Magnification in 0.001
2003 _h :1F _h (00F1) _h	Analog 10V corresponds to the speed value in rpm

Enable

There is no state machine management in this mode and the motor is enabled to run via the

terminals.

Operating prerequisites: The integrated low-voltage servomotor is switched to NiMotion speed mode (2002_h:01_h (00B1_h) set to 2).

1. Speed source selection:

Set the primary speed command source 2006_h:01_h (0148_h) and the secondary speed command source 2006_h:02_h (0149_h) to 1. Meaning: 1 - Analog speed regulation;

By means of the speed command selection 2006_h:03_h (014A_h) the actual effective is the main speed command (mode 1 analog speed control) or the auxiliary speed command (mode 2 analog speed control), meaning: 0 - main speed given, 1 - auxiliary speed given;

2. Set the multi-function terminals (both physical and virtual terminals are applicable, see "7.1 Digital Inputs and Outputs" for details on terminal settings):

Configure one input terminal as motor enable, and configure its valid logic level according to the actual situation, the default low level is valid.

If the speed command selection 2006_h:03_h (014A_h) set in the previous step is 0, you need to set the corresponding input terminal DI1 or DI2 for the analog input speed control direction function (multifunction number 36);

3. Velocity, acceleration and maximum speed settings

If the speed command selection 2006_h:03_h (014A_h) is 0, you need to configure the analog 10V corresponding to the speed value 2003_h:1F_h (00F1_h);

If the speed command selection 2006_h:03_h (014A_h) is 1, the forward and reverse speed thresholds 2006_h:0A_h (0151_h) and 2006_h:0B_h (0152_h) need to be configured;

Configure the plus and minus speeds 2006_h:07_h (014E_h) and 2006_h:08_h (014F_h);

4. run

The motor is enabled via a terminal and runs at the corresponding speed according to the analog input.

Analog speed control mode

There are two modes of analog input speed control: mode 1 and mode 2.

1. Model 1:

The given speed before dead-band = sign (corresponding to the level of multi-function number 36) * analog 10V corresponding speed value * (input voltage * AI1 multiplier + AI1 bias)/10000, multi-function number 36 is valid for negative direction, invalid for positive direction.

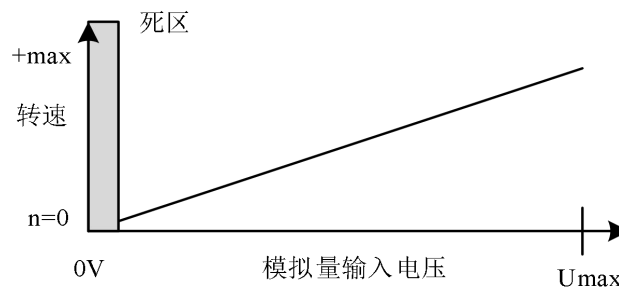


Figure 6- 3

2. Mode 2:

Input voltage 5~10V: given speed before dead-band = speed value corresponding to analog 10V * (input voltage * AI1 multiplier + AI1 bias)/10000.

Input voltage 0~5V: the given speed before dead-band = - the speed value corresponding to analog 10V * (input voltage * AI1 multiplier + AI1 bias)/10000.

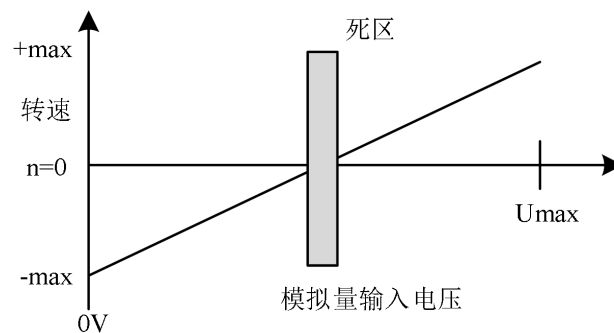


Figure 6- 4

Configuration Example

● Parameters setting

Set to NiMotion speed mode: 2002_h:01_h (00B1_h)=2;

Send: 01 10 00 B1 00 01 02 00 02

Set the main speed command source to 1: 2006_h:01_h (0148_h)=1 and set the speed command selection 2006_h:03_h (014A_h)=0;

Send: 01 10 01 48 00 01 02 00 01 (set primary speed source to 1)

Send: 01 10 01 4A 00 01 02 00 00 (set speed command selection as main speed command source)

Set the acceleration ramp time constant to 100ms: 2006_h:07_h (014E_h)=100 (in ms);

Send: 01 10 01 4E 00 01 02 00 64

Set the deceleration ramp time constant to 100ms: 2006_h:08_h (014F_h)=100 (in ms);

Send: 01 10 01 4F 00 01 02 00 64

Set the analog 10V corresponding to the speed value, 2003_h:1F_h (00F1_h)=3000 (unit rpm);

Send: 01 10 00 F1 00 01 02 0B B8

Set the DI1 and DI2 terminals:

DI1 function selected as servo enable: 2003_h:03_h (00D5_h) = 1;

Send: 01 10 00 D5 00 01 02 00 01

DI1 logic selection active low: 2003_h:04_h (00D6_h)=0;

Send: 01 10 00 D6 00 01 02 00 00

DI2 function selected for analog input speed control direction: 2003_h:05_h (00D7_h) = 36;

Send: 01 10 00 D7 00 01 02 00 24

DI2 logic select active low: 2003_h:06_h (00D8_h) = 0;

Send: 01 10 00 D8 00 01 02 00 03

The AI interface accesses the analog voltage value (be careful not to reverse the positive and negative), which is used to adjust the voltage speed regulation;

- Running:

DI1 inputs low, the motor runs at the speed given by the analog 2006_h:04_h (014B_h), and DI2 controls the direction of motor operation.

6.2.3 Analog torque control

Table 6- 11

Object Index	Description
2002 _h :01 _h (00B1 _h) _h	Operation mode selection
2007 _h :01 _h (015E _h) _h	NiMotion torque mode main torque feed source, 1 - Mode 1 analog torque control;
2007 _h :02 _h (015F _h) _h	NiMotion torque mode auxiliary torque feed source, 1-mode 2 analog torque control;
2007 _h :03 _h (0160 _h) _h	NiMotion torque mode torque command selection, 0 - primary torque given, 1 - secondary torque given;
2007 _h :04 _h (0161 _h) _h	Torque command digital given value (0.1%)
2007 _h :0A _h (0167 _h) _h	Positive internal torque limit (0.1%)
2007 _h :0B _h (0168 _h) _h	Reverse internal torque limit (0.1%)
2007 _h :10 _h (016D _h) _h	Torque control forward speed limit value (rpm)
2007 _h :11 _h (016E _h) _h	2007 _h :11 _h : Torque control reverse speed limit value (rpm)
2017 _h (0326 _h ~0346 _h) _h	Functional configuration of the virtual input terminals (see: "6.1.1 Digital input")
2003 _h :17 _h (00E9 _h) _h	AI1 bias in 1mV
2003 _h :18 _h (00EA _h) _h	AI1 input filter time constant in 0.01ms
2003 _h :19 _h (00EB _h) _h	AI1 Deadband in 1mV
2003 _h :1A _h (00EC _h) _h	AI1 Magnification in 0.001
2003 _h :20 _h (00F2 _h) _h	Analog 10V corresponds to the torque value, unit 0.1%.

Enable

There is no state machine management in this mode and the motor is enabled to run via the terminals.

Operating prerequisites: The integrated low-voltage servomotor is switched to NiMotion torque mode (2002_h:01_h (00B1_h) set to 3).

1. Torque source selection:

Set the primary torque command source 2007_h:01_h (015E_h) and the secondary torque command source 2007_h:02_h (015F_h) to 1. Meaning: 1 - Analog torque control;

By means of the torque command selection 2007_h:03_h (0160_h) the actual effective is either a main torque command (mode 1 analog torque control) or an auxiliary torque command (mode

2 analog torque control), meaning: 0 - main torque given, 1 - auxiliary torque given;

2. Set the multi-function terminals (both physical and virtual terminals are applicable, see "6.1 Digital Inputs and Outputs" for details on terminal settings):

Configure one input terminal as motor enable, and configure its valid logic level according to the actual situation, the default low level is valid.

If the torque command selection $2007_h:03_h$ (0160_h) set in the previous step is 0, you need to set the corresponding input terminal DI1 or DI2 for analog input torque control direction function (multifunction number 36);

3. Torque setting and torque and speed limit settings

If the torque command selection $2007_h:03_h$ (0160_h) is 0, you need to configure the analog 10V corresponding to the torque value $2003_h:20_h$ ($00F2_h$);

If the torque command selection $2007_h:03_h$ (0160_h) is 1, you need to configure the positive and negative internal torque limit values $2007_h:0A_h$ (0167_h) and $2007_h:0B_h$ (0168_h);

Configure the forward and reverse speed limit values $2007_h:10_h$ ($016D_h$) and $2007_h:11_h$ ($016E_h$);

4. run

The motor is enabled via one terminal and the motor is controlled according to the analog input.

Analog torque control mode

The analog input torque control has two modes: mode 1 and mode 2.

1. Model 1:

Given torque = Symbol (Multi-function No. 36 level corresponds to) * Analog 10V corresponds to torque value * (Input voltage * AI1 multiplier + AI1 bias)/10000, Note: Multi-function No. 36 is valid for negative direction, invalid for positive direction.

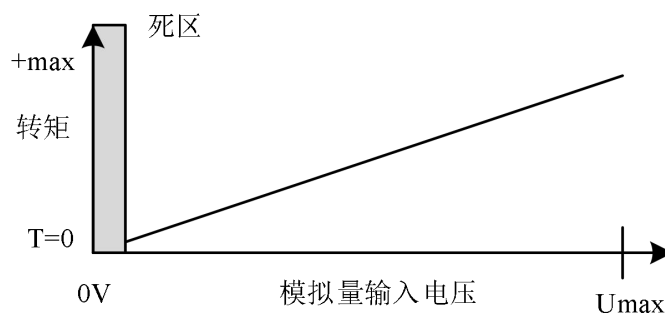


Figure 6- 5

2. Mode 2:

Input voltage 5~10V: given torque = Symbol (multifunction number 36 level corresponds to) * Analog 10V corresponding torque value * (Input voltage * AI1 multiplier + AI1 bias)/10000;
Input voltage 0~5V: given torque = -symbol (multifunction number 36 level corresponds to) * analog 10V corresponds to torque value * (input voltage * AI1 multiplier + AI1 bias)/10000;

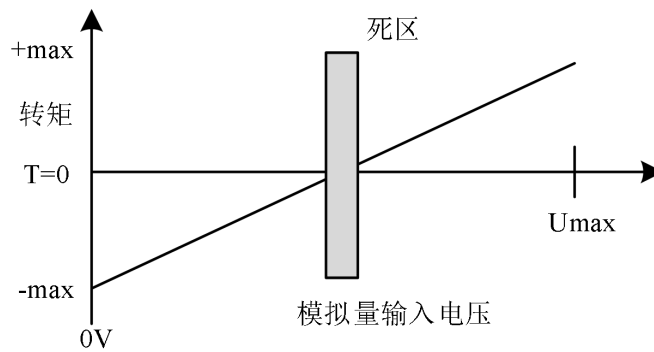


Figure 6- 6

Configuration Example

- Parameters setting

Set to NiMotion torque mode: 2002_h:01_h (00B1_h)=3;

Send: 01 10 00 B1 00 01 02 00 03

Set the main torque command source to 1: 2007_h:01_h (015E_h)=1;

Send: 01 10 01 5E 00 01 02 00 01

Setting torque command selection is for the main torque command: 2007_h:03_h (0160_h)=0;

Send: 01 10 01 60 00 01 02 00 00

Set the positive and negative internal torque limits to 100%: 2007_h:0A_h (0167_h)=1000 and 2007_h:0B_h (0168_h)=1000 (unit 0.1%);

Send: 01 10 01 67 00 01 02 03 E8 (sets positive internal torque limit to 100%)

Send: 01 10 01 68 00 01 02 03 E8 (sets reverse internal torque limit to 100%)

Set the forward and reverse speed limit values to 3000 rpm: 2007_h:10_h (016D_h)=3000 and 2007_h:11_h (016E_h)= 3000 (in rpm);

Send: 01 10 01 6D 00 01 02 0B B8 (set forward speed limit value to 3000rpm)

Send: 01 10 01 6E 00 01 02 0B B8 (set reverse speed limit value to 3000rpm)

Set the analog 10V corresponding to the torque value, 2003_h:20_h (00F2_h)=1000 (unit 0.1%);

Send: 01 10 00 F2 00 01 02 03 E8

Set the DI1 and DI2 terminals:

DI1 function selected as servo enable: 2003_h:03_h (00D5_h) = 1;

Send: 01 10 00 D5 00 01 02 00 01

DI1 logic select active low: 2003_h:04_h (00D6_h) =0;

Send: 01 10 00 D6 00 01 02 00 00

Set DI2 function selection to analog input position control direction: 2003_h:05_h (00D7_h)=36, logic to falling edge valid: 2003_h:06_h (00D8_h)=3;

Send: 01 10 00 D7 00 01 02 00 24 (set DI2 to analog input position control direction 36)

Send: 01 10 00 D8 00 01 02 00 03 (sets DI2 logic to be active on falling edge)

- run

DI1 input is low, the motor runs according to the torque given by the analog 2007_h:04_h (0161_h), and DI2 controls the direction of motor operation.

6.3 Holding Brake Settings

A holding brake is used to prevent the motor shaft from moving when the motor is in a non-enabling state, thus locking the motor in position. The holding brake is categorized into two ways: external holding brake and internal holding brake. The integrated motor automatically switches the state of the holding brake output according to the CiA402 state machine as shown below:

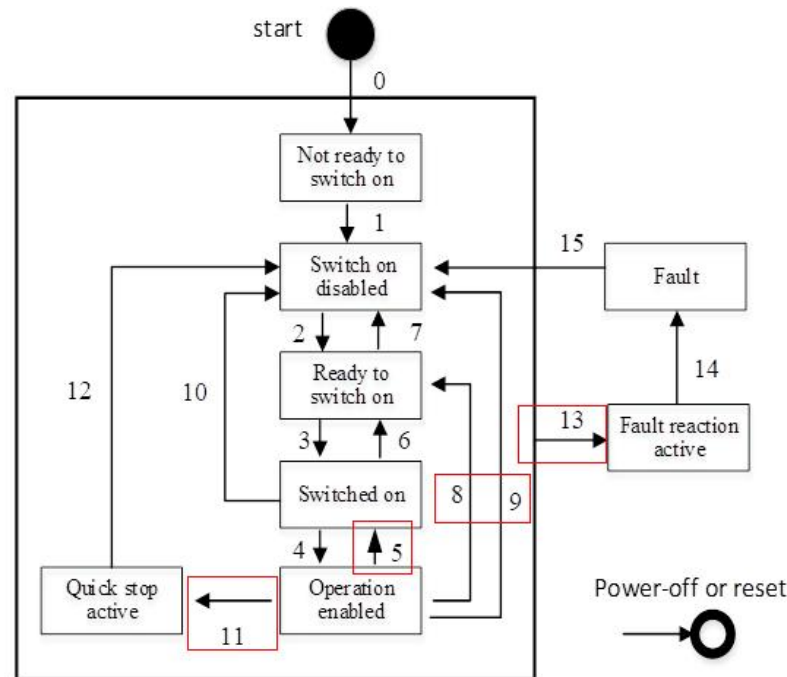


Figure 6- 7

Note: When the state machine leaves the operation state (Operation enable), the holding brake output is closed.

Table 6- 12

Bit	Name	Description
0	Motor ready.	When both status words Bit0- bit3 are 1, the holding brake output is turned on.
1	Turn on the main circuit power	
2	Quick Stop	
3	Motor operation	

Table 6- 13

Object Index	Description
2002 _h : 0A _h (00BA) _h	Holding brake output off to motor not energized delay time, unit: ms
200A _h : 1F _h (01D8) _h	Brake detection enable, 0-no gravity preset, 1-enable gravity preset
200A _h : 20 _h (01D9) _h	Holding brake load gravity (gravity preset value, when the holding brake is open, the electric machine output torque), 0.1% rated torque
200D _h : 0B _h (0254) _h	Holding brake load recognition, 0-does not enable gravity recognition, 1-enables in the linear gravity recognition

Notes:

1. When the motor is used in vertical axis, it is necessary to turn on the gravity preset and online gravity recognition function, through the gravity preset value to prevent the load from self-weight or external force caused by the load.

into the holding brake release and load drop during motor startup.

2. when the motor is used for vertical axis, it is necessary to set the holding brake output to motor not energized delay time appropriately, to prevent the motor from running to stop or fault stopping when the

Load drops due to self-weight or external forces.

3. Enable online real-time identification of load gravity and automatically update the gravity preset value 200A_h:20_h (01D9_h) when the motor speed is below 300rpm.

Enable and disable

a) The holding brake output control can be enabled via the object dictionary 2004_h:01_h (00F8_h) and the holding brake function is automatically enabled when its function number is set to 30; When the terminal function 2004_h:01_h (00F8_h) of DX1 is configured with multifunction No. 5 (motor holding state output) and the logic is active high, DX1 outputs a high level when the holding brake is opened.

b) The hold output control can be switched off using the object dictionary 2004_h:01_h (00F8_h), and the hold function is automatically switched off when its function number is set to 0. When the terminal function 2004_h:01_h (00F8_h) of DX1 is configured with multifunction No. 5 (motor holding state output) and the logic is active high, DX1 outputs a low level when the holding brake is closed.

duty cycle

The brake outputs trigger a PWM signal with adjustable duty cycle. The duty cycle (ratio of pulse time to cycle time) is set in 2002_h:04_h (00B4_h) and is a percentage value that can be selected between 1 and 100 (a configuration above 90 is recommended). When the value is 100, the brake output will be on continuously.

6.4 I² t Overload Protection of Motor

After an integrated low-voltage servo motor is energized, it will continuously generate heat due to the thermal effect of the current, and at the same time release heat to the surrounding environment. When the heat generated by the motor exceeds the heat released, the motor temperature rises, and an excessively high temperature will cause the motor to burn out. I² t The purpose of the motor overload protection is to ensure that the motor operates normally under the temperature limit and to prevent damage to the motor.

relevant object

Table 6 - 14

Object Index	Description
2000 _h :09 _h (0065 _h)	Rated current (unit: 0.01A)
2000 _h :0A _h (0066 _h)	Maximum current (unit: 0.01A)
2000 _h :0B _h (0067 _h)	Maximum current duration (unit: 0.1s)